

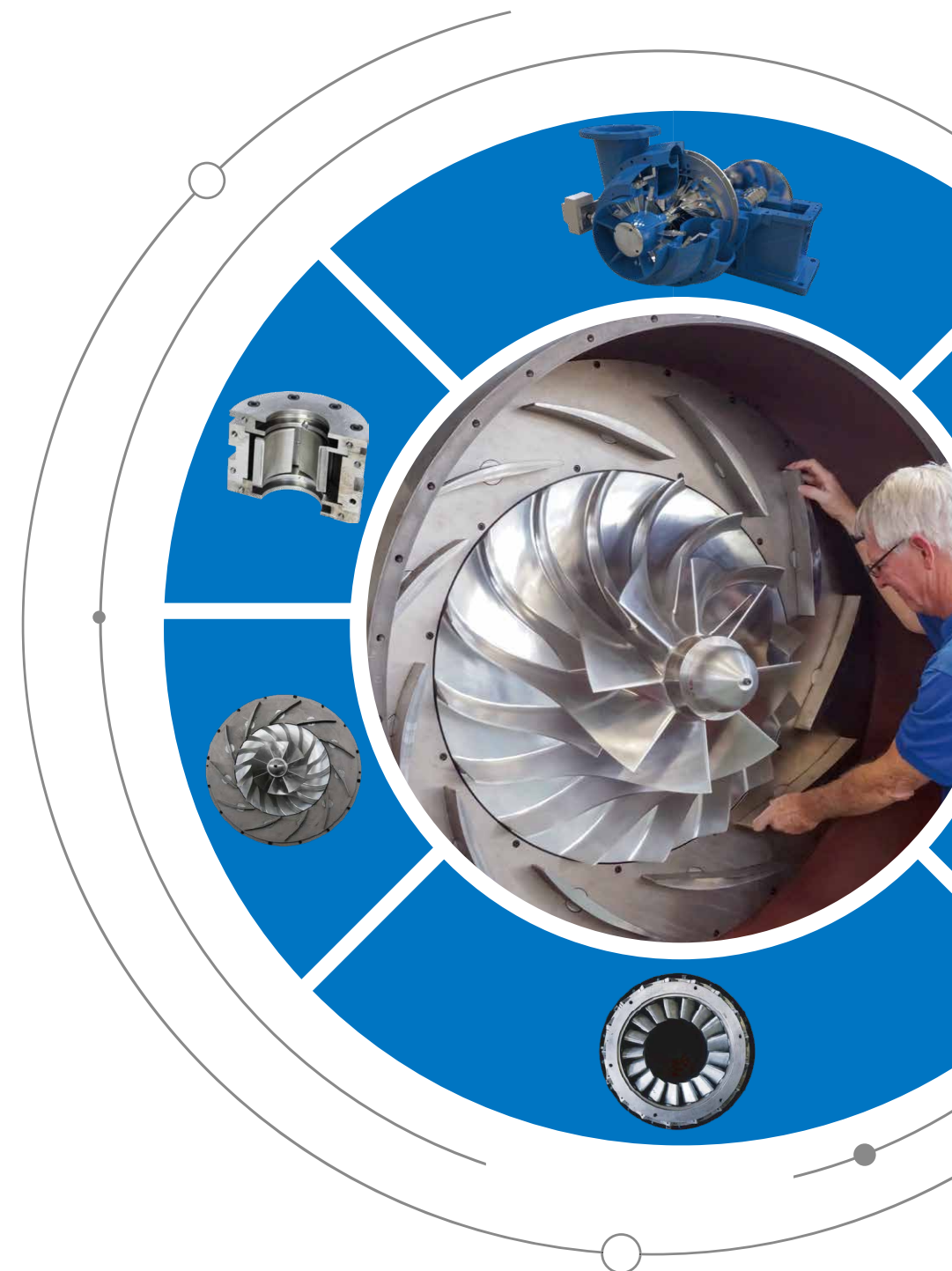
- ★ GL-TURBO Global Headquarters
- GL-TURBO Global Branches



GL-TURBO
USA · China · Denmark · Germany · India

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GL-TURBO (2401)_US_Headquarter 04 | 2024



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About GL-TURBO®

GL-TURBO® was founded in Boston USA in 2006 to disrupt the geared turbo market long dominated by a couple of international heavyweights since the 1980s.

To date, GL-TURBO® has established subsidiaries in Houston, USA; Leipzig, Germany; Copenhagen, Denmark; Wuxi, China and Bengaluru, India, fully leveraging local advantages and employing the most rational methods for R&D, production, sales, and after-sales service. GL-TURBO® single-stage centrifugal blowers, GPAC® precision control systems, and related products have served more than 200 customers in over 20 major countries worldwide.



History



Key Technologies

GL-TURBO® single-stage centrifugal blowers possess comprehensive intellectual property rights. The company's founders and core technical staff hail from globally renowned institutions such as the Massachusetts Institute of Technology (MIT), General Electric Company (GE), the former Northern Research and Engineering Corporation (NREC), the Technical University of Denmark (DTU), Shanghai Jiao Tong University (SJTU), and others, bringing together more than 200 years of collective expertise in the design and manufacturing of blowers and compressors.

Impeller and Aerodynamic Design

GL-TURBO®'s high efficiency comes from the latest aerodynamic design in the industry. All aerodynamic components, including impellers, shrouds, volutes, inlet guide vanes, and outlet guide vanes, are integrated system.

For every blower model, GL-TURBO® offers a choice between two distinct aerodynamic designs, thereby guaranteeing optimal design efficiency across a wide range of operating conditions.

Tri-flow Impeller

Large backsweep

- Highest aerodynamic efficiency



Small backsweep

- Smaller impeller
- Suitable for high pressure
- Slightly lower aerodynamic efficiency



Adjustable Inlet Guide Vanes

By utilizing adjustable inlet guide vanes to pre-swirl the airflow before it enters the impeller, both flow and pressure conditions are optimized, ultimately resulting in enhanced impeller efficiency. In comparison to using valves for inlet airflow adjustment, manipulating the angle of the inlet guide vanes can effectively reduce power loss by 5% to 15%.



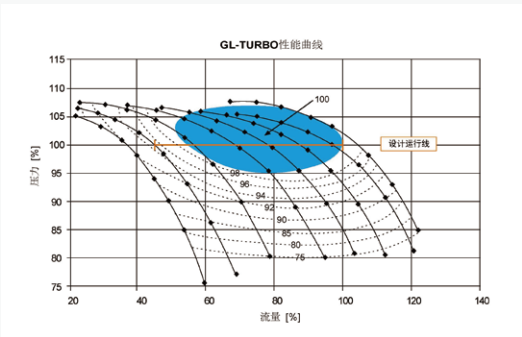
Adjustable variable Vane Diffuser

Adjustable variable Vane Diffuser enables the matching of various flow rates and pressure ratios, thereby maximizing the unit's overall efficiency. The diffuser vanes feature a streamlined, asymmetric cross-sectional design, which allows for an adjustable flow area while simultaneously minimizing air outlet resistance and maintaining an adequate surge margin.



Dual Guide Vane Adjustment

GL-TURBO® single-stage centrifugal blowers attain the widest flow regulation range, spanning from 40% to 100%, by employing a coordinated adjustment of both inlet and outlet guide vanes. This innovative design allows the blowers to maintain high efficiency across the entire regulation range, ultimately achieving unparalleled overall operating efficiency.



Overall Efficiency

In most cases, the blower flow rate must be adjusted to accommodate varying process requirements. Consider a scenario where, over the course of a year, the process demands flow rates at 100%, 80%, 60%, and 40% of the rated flow rate, with corresponding operating times of 10%, 40%, 40%, and 10%, respectively. In this situation, a GL-TURBO® single-stage centrifugal blower with a rated shaft power of 200kW can achieve energy savings of at least 11% or more when compared to a conventional single-stage centrifugal blower lacking adjustable inlet and variable Vane Diffuser, which can only be turned down to a minimum flow rate of 75%.

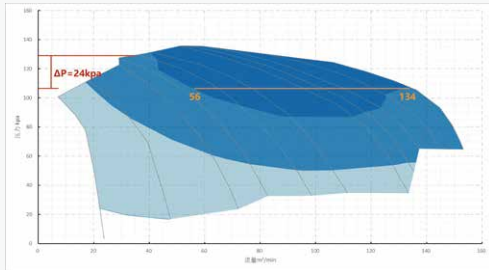
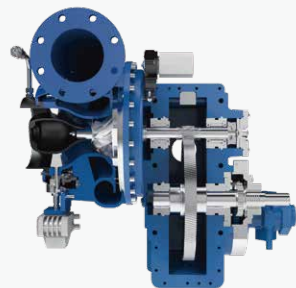
Flow	Operating Time	Blower Power Consumption (kW)		GL-TURBO Energy Saving (kW)	
		GL-TURBO	Typical Brand	Difference	Time-weighted
100%	10%	208.33	219.30	10.96	1.10
80%	40%	166.67	175.44	8.77	3.51
60%	40%	125.00	164.47	39.47	15.79
40%	10%	114.58	164.47	49.89	4.99
Overall saving - Unit (kWh)					25.38
Energy Saving Ratio					11.58%
Blower Power Consumption (kWh)					222,362

Product Series

GL-TURBO® customizes its single-stage centrifugal blowers to optimally match the specific operating conditions of customers across the globe. The company offers three distinct series: GL, GA, and DT, each designed to cater to diverse customer needs in terms of stability, operating efficiency, flow regulation, and precise control.

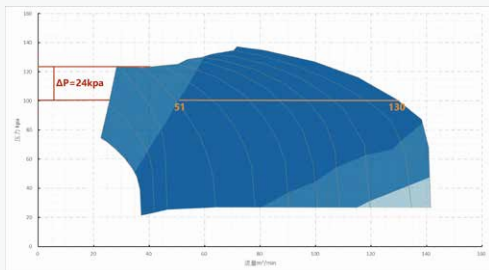
GL Series Geared Turbo blower

- Inlet guide vanes + Variable Vane Diffuser
- Geared
- Balanced stability and overall operating efficiency
- Maximum power 4500kW



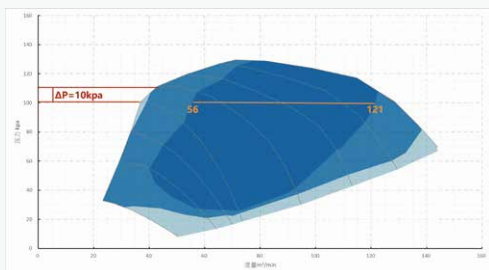
GA Series GA Hybrid Turbo Blower

- Inlet guide vanes + Variable Vane Diffuser
- Permanent magnet synchronous motor + frequency converter
- Combines the advantages of geared and PM motor direct drive



DT Series DT Maglev Centrifugal Blower

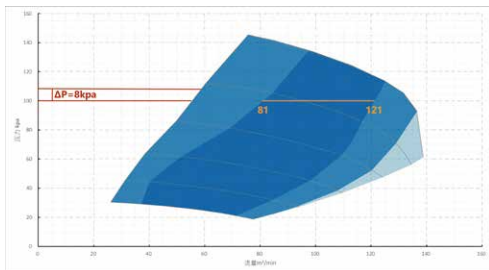
- Inlet guide vanes
- Permanent magnet synchronous motor + frequency converter
- Balanced stability and overall operating efficiency
- Highest overall operating efficiency



GL-TURBO® offers superior stability, a wider surge margin, and an expanded high-efficiency operating range.

Typical high-speed motor direct drive
Air suspension and magnetic suspension blowers

- Only frequency conversion
- No guide vane control
- Small high-efficiency zone
- Small surge margin



Manufacturing and Testing Standards

Manufacturing Standards

GL-TURBO®'s dedication to quality and environmental stewardship encompasses every facet of its operations, from design and manufacturing to testing and service. The company adheres to the stringent requirements set forth by all major international standards, including:

- API 672
Packaged, Integrally Geared Centrifugal Air Compressors for Petroleum, Chemical, and Gas Industry Services.
- ASME PTC-10 (1997)
Performance Test Code on Compressors and Exhausters
- ASME PTC-13 (2018)
【GL-TURBO participated in the formulation】
Wire-to-Air Performance Test Code for Blower Systems
- ISO 9001:2015
Quality Management System Certification
- ISO14001:2015
Environmental Management System Certification

Quality Inspection

GL-TURBO® rigorously tests every blower in strict accordance with the latest international standard, ISO 13350, prior to shipment.

- Dynamic balance test
- No-load test
- Load test
- Overspeed test
- Efficiency test



▲ Impeller inspection ▲ Gear inspection ▲ Dynamic balance test

Performance Guarantee

Testing can be conducted at either the Houston factory or the Wuxi factory, in full compliance with ASME PTC10, ASME PTC13, and ISO 5389 standards

We offer comprehensive certification documents, state-of-the-art facilities, and a highly skilled professional team. Customers are welcome to observe any stage of the testing process to verify that the blower performance meets the most stringent standards.



Service

Beyond delivering stable, efficient blowers and precise control systems, GL-TURBO® is committed to providing customer-centric, comprehensive pre-sales, in-sales, and after-sales services.

Pre-sales Service

Prior to the sale, GL-TURBO® equips customers with in-depth knowledge about blowers and precise control systems. This enables customers to select the appropriate product series, models, and related auxiliary equipment for the precise control system, customeried to their specific application requirements.



In-sales Service

Throughout the sales process, GL-TURBO® offers customers installation guidance for blowers and precise control systems, as well as testing of instruments and meters. Additionally, the company provides training for maintenance technician to ensure stable operation and optimal performance.



After-sales Service

In every major country worldwide, GL-TURBO® has a network of experienced technical and rigorously trained partners who deliver swift and professional after-sales services to customers. These services include regular maintenance, remote monitoring, fault diagnosis and recovery, and spare parts provision, ensuring optimal performance and minimizing downtime.



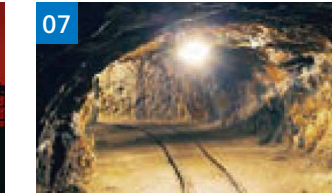
Upon receiving a customer service request GL-TURBO® guarantees a response within 24 hours, demonstrating its commitment to prompt and attentive customer support.

Spare Parts

GL-TURBO® maintains permanent records of all blowers, including electronic documents and written files. The company is committed to providing all necessary spare parts for blowers post-sale. Typical spare parts include:

 Tilting pad bearings	 Hydrodynamic bearing	 High-efficiency impellers	 Air filter elements
 Labyrinth seals	 Bearing oil seals	 Variable Vane Diffuser actuators	 Inlet guide vane actuators

Application

 01 Water And Sewage	 02 Biogas	 03 Vacuum And Pneumatic Conveying	 04 Papermaking
 05 Petrochemical	 06 Power	 07 Mining	 08 Textile
 09 Yeast Fermentation	 10 Seawater Desalination	 11 Glass	 12 Food And Beverage